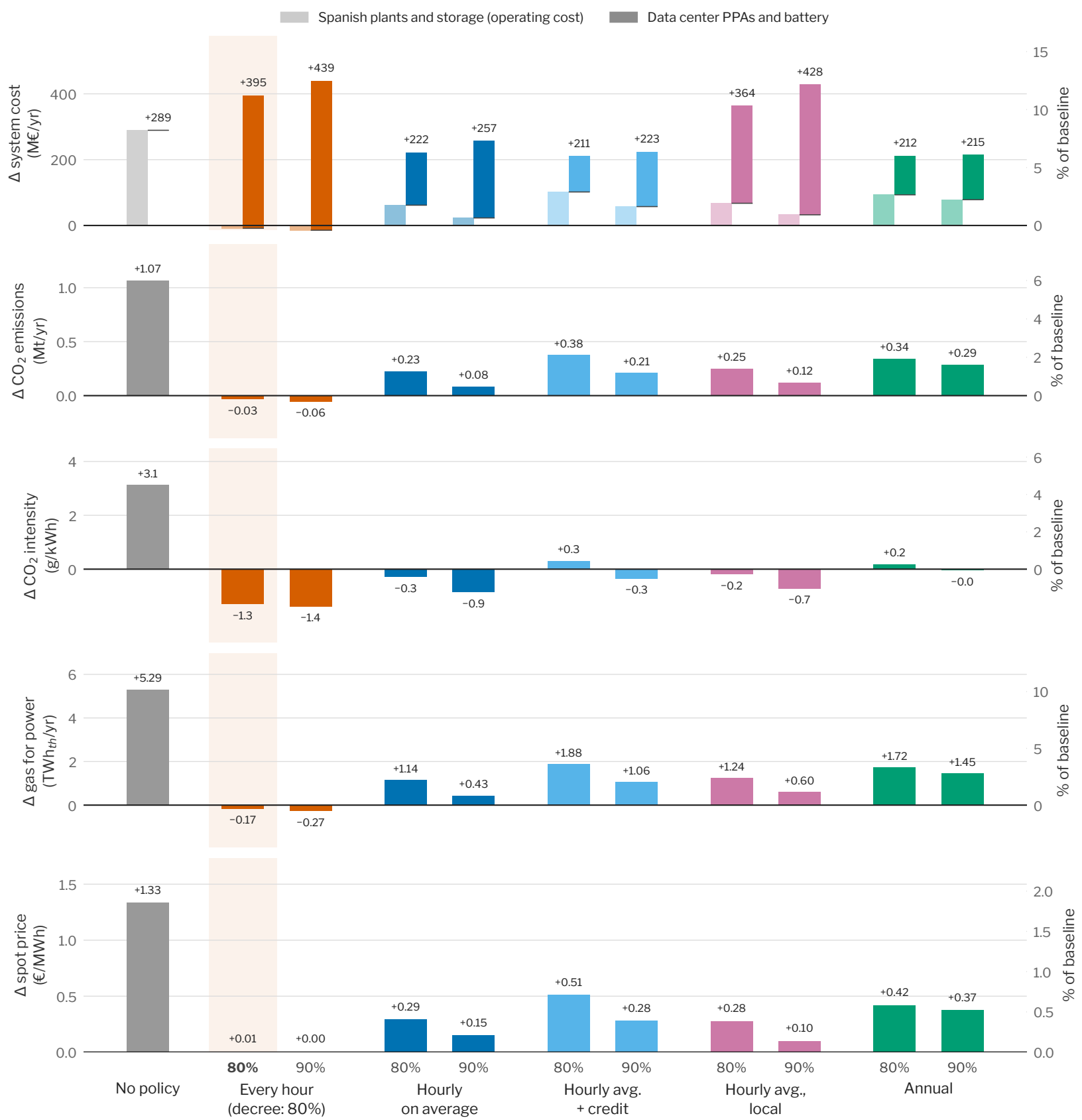

Change in the Spanish power system with a 500 MW data center in Aragon against a baseline without it, absolute (left axes) and in percent of the baseline level (right axes). System cost: operating cost of Spanish plants and storage (light), then the data center's PPA and battery costs on top (dark), ending at the total. CO2: fuel burnt in Spanish plants times its carbon content. CO2 intensity: Spanish CO2 per kWh of Spanish generation, the data center's PPAs included. Gas: fuel burnt in Spanish gas-fired power plants (CCGT, OCGT). Spot price: hourly nodal prices weighted by Spanish demand, the data center excluded. PPAs anywhere in mainland Spain unless marked local (Aragon only).